Practical - 11

Practical Title: Student Management CRUD with Search and Filter Problem definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

A. Theory:

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

   Answer: Yes, all CRUD operations are working correctly. SELECT displays student records, INSERT adds new students, UPDATE modifies student details, and DELETE removes student records.

2. Is search by name/email/course implemented?

   Answer: Yes, search by student name, email, and course is implemented. The course filter also displays students from the selected course.

3. Are delete operations confirmed before execution?

   Answer: Yes, a confirmation message appears before deleting a student record. The record is deleted only after clicking OK.

A. Implementation

PHP Functions and MySQLi Methods Used

Function / Method	Purpose
new mysqli()	Connects PHP to MySQL.
prepare()	Prepares an SQL statement.
bind_param()	Binds values to query parameters.
execute()	Executes the SQL statement.
get_result()	Retrieves query results.
fetch_assoc()	Fetches a row as an associative array.
trim()	Removes extra whitespace from input.
htmlspecialchars()	Helps safely display user-provided text.
confirm()	Asks for confirmation before deletion.
isset()	Checks whether a variable or form value exists.

Code	Purpose
$conn = new mysqli(...)	Connects PHP to MySQL
INSERT INTO	Adds a new student
SELECT	Retrieves student records
UPDATE	Modifies student information
DELETE FROM	Removes a student
$conn->prepare()	Prepares an SQL statement safely
bind_param()	Binds values to the query parameters
LIKE	Searches for matching text
LIMIT and OFFSET	Display records page by page
confirm()	Asks the user before deletion
htmlspecialchars()	Helps prevent HTML/script injection when displaying data
filter_var()	Validates the email format

B. Evidence of Work (Screenshots)

Student Management System

+ Add Student

Search name, email or cour | All Courses | Search / Filter | Reset

ID	Name	Email	Course	Actions
20	Tara Joshi	tara@example.com	BCA	Edit Delete
19	Neil Desai	neil@example.com	BTech	Edit Delete
18	Pari Mehta	pari@example.com	BSc IT	Edit Delete
17	Aditya Patel	aditya@example.com	BCA	Edit Delete
16	Myra Shah	myra@example.com	BTech	Edit Delete
15	Yash Joshi	yash@example.com	BSc IT	Edit Delete
14	Kiara Desai	kiara@example.com	BCA	Edit Delete
13	Dev Mehta	dev@example.com	BTech	Edit Delete
12	Riya Patel	riya@example.com	BSc IT	Edit Delete
11	Kabir Shah	kabir@example.com	BCA	Edit Delete
10	Mira Joshi	mira@example.com	BTech	Edit Delete
9	Krish Desai	krish@example.com	BSc IT	Edit Delete

Student Management System

+ Add Student

Sara | All Courses | Search / Filter | Reset

ID	Name	Email	Course	Actions
8	Sara Mehta	sara@example.com	BCA	Edit Delete

C. GitHub Repository Link:

https://github.com/heerce27/StudentHub-Portal/tree/ed9b309f7c40e08097f20a8c86c3b8f7277bc24a/practical11

D. Challenges Faced:

1. Performing INSERT, SELECT, UPDATE, and DELETE operations correctly.
2. Implementing search by name, email, and course.
3. Displaying proper success and error messages.

Learning Outcomes:

1. Understood how to perform CRUD operations on student records.
2. Learned to search students by name, email, and course.
3. Understood how to filter records based on the selected course.

Faculty Remarks: